

Benign, Harmful or Novel? Triaging Concept Drift for In-Context Tabular Learners

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Abstract

Tabular foundation models (TFMs) adapt to a data stream by changing their context rather than their weights. We ask which concept drifts actually harm an in-context learner and which of them it can see. On controlled streams with TabPFN, a shift of the input distribution inside the support of the context costs no accuracy even when a two-sample test separates old from new rows with AUROC 0.95, a shift outside the support costs as much as a change of the labelling function, and a change of the labelling function leaves the model’s confidence untouched. We prove that no label-free statistic can detect such a real drift, that covered drift is benign, and we propose DRIFT-TRIAGE, which triages the error stream of a standard drift detector by coverage and class novelty. In a pre-registered held-out evaluation it matches the accuracy of a detector-triggered reset, cuts needless resets from 47 to 9 and keeps the detection of new classes outside the support (AUROC 0.79 versus 0.61), while unsupervised drift detectors fire on benign drift and miss harmful drift.

1. Introduction

Momentum. Prior-data fitted networks approximate Bayesian posterior prediction in one forward pass (Müller et al., 2022; Nagler, 2023), and TFMs such as TabPFN (Hollmann et al., 2025) and TabICL (Qu et al., 2025) now classify tables without gradient updates. Because they learn from the context, they are a natural fit for data streams, where the context is a sketch of recent data (Lourenço et al., 2026), and for temporal shift (Helli et al., 2024).

Problem. Streams are non-stationary: their joint distribution changes over time (Gama et al., 2014; Lu et al., 2019). Concept drift theory separates *virtual* drift of $P(\mathbf{x})$ from

real drift of $P(y | \mathbf{x})$ and treats the appearance of an unseen class as a sudden drift. For an in-context learner these types have different consequences, but existing TFM work treats drift as one phenomenon and measures only accuracy after adaptation.

Observation 1. The type of a drift, not the size of the input shift, decides whether a TFM fails. When $P(\mathbf{x})$ is tilted inside the support of the context, a domain classifier separates old from new rows with AUROC up to 0.95, yet the stale context loses at most 0.006 accuracy, within seed noise. When the labelling function changes on the same inputs, the loss equals the fraction of rows whose label changed, within 0.021 (hyperplane rotated by 90°: 0.51 versus 0.98). When $P(\mathbf{x})$ moves outside the support, the stale context fails as well (periodic boundary translated by twice its range: 0.52 versus 0.99). (Held-out seeds; the pre-registered bound of 0.005 for the tilt was missed by 0.0006.)

Observation 2. Label-free signals see the wrong drifts. Under real drift the TFM’s confidence does not move (it changes by at most 0.004; 0.974 at 90°, where accuracy is 0.51) and the input distribution is unchanged, so neither confidence nor a $P(\mathbf{x})$ detector can see it; a $P(\mathbf{x})$ detector instead fires on the benign in-support tilt. Uncovered inputs, in contrast, lower the confidence (0.66) and are far from the context: a nearest-neighbour coverage ratio is ≤ 1.07 for every in-support tilt and ≥ 3.6 for every translation.

Together these observations split concept drift for in-context learners along two axes: whether the drift harms the in-context predictor, and whether it is visible without labels. This raises the question: *how should an in-context learner use the signals it has, label-free coverage and delayed labels, to keep its context when drift is benign and change it only when drift is harmful?*

Approach. We formalise concept drift for in-context learners (Section 2) and decompose the discrepancy between the context and an incoming window into a density ratio $r(\mathbf{x})$ and a likelihood ratio $v(\mathbf{x}, y)$, following the sampling-transfer view of I-Div (Zhao et al., 2024). In practice the two signals an in-context learner has are label-free coverage, which reveals uncovered drift, and delayed labels, which alone can reveal real drift (Proposition 1). DRIFTTRIAGE

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therefore triages the error stream: errors on uncovered rows extend the context, errors on rows of a new class are rejected, and only the remaining errors can trigger a reset.

Contributions. (1) A definition of concept drift for in-context learners that separates benign, harmful and novel drift (Definition 1). (2) Observations showing that the drift type, not its magnitude, decides whether a TFM fails, and that TFM confidence cannot tell the types apart. (3) DRIFT-TRIAGE, a training-free triage of the error stream, non-inferior in accuracy to the strongest detector while avoiding needless resets and preserving novelty detection, with propositions on what is and is not identifiable. (4) A benchmark of tabular streams with typed drift events and pre-registered held-out evaluation.

2. Concept Drift for In-Context Learners

Let a stream $\mathcal{S}_{0,t} = \{(\mathbf{x}_i, y_i)\}$ arrive in windows $W_1, W_2, \dots$ and let $\mathcal{C}_t$ be the context of a frozen TFM $f_{\Theta}(\cdot; \mathcal{C}_t)$ at time t , a subset of the labelled rows seen so far. Following Lu et al. (2019), concept drift occurs at t if $P_t(\mathbf{x}, y) \neq P_{t+1}(\mathbf{x}, y)$, with $P_t(\mathbf{x}, y) = P_t(\mathbf{x}) P_t(y | \mathbf{x})$.

Definition 1 (In-context concept drift). Let $P_{\mathcal{C}}$ be the distribution of the context and P_W that of an incoming window, with label sets $\mathcal{Y}_{\mathcal{C}}$ and $\mathcal{Y}_W$. Write $r(\mathbf{x}) = P_W(\mathbf{x})/P_{\mathcal{C}}(\mathbf{x})$ and $v(\mathbf{x}, y) = P_W(y | \mathbf{x})/P_{\mathcal{C}}(y | \mathbf{x})$ for $y \in \mathcal{Y}_{\mathcal{C}}$. The drift from $\mathcal{C}$ to W is

- *covered* (benign virtual) if $v \equiv 1$ and r is bounded on the support of P_W ;
- *uncovered* (coverage drift) if $v \equiv 1$ but P_W puts mass where $P_{\mathcal{C}}$ has none (r unbounded);
- *real* if $v \not\equiv 1$ on a set of positive P_W -measure;
- *novel* if $P_W(y \notin \mathcal{Y}_{\mathcal{C}}) > 0$.

Covered drift is harmless to an in-context learner and uncovered drift is visible without labels; real drift is harmful and invisible without labels; novel rows must be rejected rather than used to change the context (Observations 1, 2).

Remark 1 (Identifiability). Real drift with an unchanged $P(\mathbf{x})$ cannot be detected from unlabelled rows alone. We therefore work in the standard prequential setting with delayed labels (Gama et al., 2014): the labels of window W_t become available after its rows are predicted, and v is estimated on the most recent labelled window, while r and novelty are estimated on the current, unlabelled one.

3. Method

DRIFTTRIAGE runs on a frozen TFM with a context of at most M rows, batches of B rows and labels one batch late.

Coverage. For each incoming (unlabelled) row, the coverage ratio is its nearest-neighbour distance to the context divided by the mean leave-one-out nearest-neighbour distance within the context; a row is *uncovered* if the ratio exceeds $\tau = 3$. Uncovered rows, once labelled, enter the context with priority: when the context is full, covered rows are evicted first.

Filtered detection. The DDM rule (Gama et al., 2014) runs only on the errors of rows that were covered and whose class is already in the context. Errors on uncovered rows signal coverage drift and errors on rows of a new class signal novelty; neither resets the context. A detection resets the context to the newly labelled rows.

Rejection. Every incoming row gets a rejection score, the larger of one minus the TFM confidence and the excess coverage ratio $\max(0, \text{ratio}/\tau - 1)$.

The only constant chosen on development streams is τ ; the DDM rule keeps its standard thresholds. The design history (two rejected variants based on context-swap tests, and two rejected refinements) is reported in the appendix. (Section A).

4. Theory

Let $\mathcal{C}$ be the context, W an incoming window, $h = f_{\Theta}(\cdot; \mathcal{C})$ the in-context predictor and $R_P(h) = \Pr_{(\mathbf{x}, y) \sim P}(h(\mathbf{x}) \neq y)$ its error under P .

Proposition 1 (Real drift is invisible without labels). *Let T be any statistic of the context and the unlabelled rows of W . If two window distributions P_W and P'_W have the same input marginal, $P_W(\mathbf{x}) = P'_W(\mathbf{x})$, then T has the same law under both. In particular, no label-free detector, including the confidence of h and any test of $P(\mathbf{x})$, has power above its level against a real drift that leaves $P(\mathbf{x})$ unchanged.*

Proof. T is a measurable function of $(\mathcal{C}, \mathbf{X}_W)$, and the law of $\mathbf{X}_W$ is the product of the input marginal, which is the same under P_W and P'_W ; $\mathcal{C}$ is fixed. Hence the laws of T coincide. $\square$

Proposition 2 (Covered drift is benign). *If the drift from $P_{\mathcal{C}}$ to P_W is covered in the sense of Definition 1, with $v \equiv 1$ and $r \leq \bar{r}$ on the support of P_W , then $R_{P_W}(h) \leq \bar{r} R_{P_{\mathcal{C}}}(h)$.*

Proof. With $v \equiv 1$ the conditional label law is the same, so the error of h at $\mathbf{x}$ is the same function $e(\mathbf{x}) = \Pr(h(\mathbf{x}) \neq y | \mathbf{x})$ under both, and $R_{P_W}(h) = \mathbb{E}_{P_{\mathcal{C}}}[r(\mathbf{x})e(\mathbf{x})] \leq \bar{r} R_{P_{\mathcal{C}}}(h)$. $\square$

Proposition 2 explains the first half of Observation 1: a TFM that is accurate on its context stays accurate under any tilt with a bounded density ratio, however large the shift

whether I-Div's
ood-ratio step
a label-free
under an ex-
umption (e.g.
hif).

Table 1. Held-out streams (seeds 10–19). Acc: prequential accuracy; Real: accuracy in the 10 batches after a real drift; Needless: resets in the 10 batches after a tilt, a translation or a novel class (sum over 20 streams); Nov: AUROC for rows of a new class before their labels arrive (sine2d).

Policy	Acc sine2d	Acc lin	Real	Needless	Nov
FIFO	0.968	0.952	0.731	0	0.762
Dual memory	0.969	0.948	0.738	0	0.763
$P(\mathbf{x})$ reset	0.967	0.950	0.731	69	0.729
DDM reset	0.978	0.969	0.913	47	0.606
DRIFTTRIAGE	0.980	0.969	0.912	9	0.786

measured by a two-sample test. When P_W puts mass where P_C has none, $\bar{r} = \infty$ and nothing is guaranteed; under a kernel posterior predictive the confidence then reverts to the prior far from the context, which makes uncovered drift visible without labels. Proposition 1 explains the second half of Observation 2, and it is why DRIFTTRIAGE detects real drift from delayed labels only.

Proposition 3 (Filtered detection). *Suppose that between two resets there is no real drift on covered rows of known classes. Then the errors of h on the rows that DRIFTTRIAGE passes to the detector (covered rows of known classes) are independent with mean at most $\bar{r} R_{P_C}(h)$, whatever coverage drift or novel classes occur meanwhile, so the false-alarm guarantee of the detector (Gama et al., 2014) applies to them.*

5. Experiments

Setup. Streams of two families with typed drift events every 3000 rows, two cycles over {tilt, translate, real, novel} in random order: *sine2d* (periodic boundary; the novel class is a band outside the support) and *lin* (four linear classes; the novel class takes over part of the support). Batches of 100 rows, labels one batch late, a context of 1000 rows, TabPFN v2. Baselines: a FIFO context, the dual memory of Lourenço et al. (2026), a context reset by DDM (Gama et al., 2014) on the error stream, and a reset by an unsupervised $P(\mathbf{x})$ detector (two-sample classifier). All design choices were made on development seeds 0–4; the hypotheses below were written down before the single run on seeds 10–19.

Results (Table 1). Detection-driven resets dominate: after a real drift, FIFO and the dual memory recover to 0.73–0.74 within ten batches, a DDM reset to 0.91, because a TFM relearns a concept from a few hundred rows. The unsupervised $P(\mathbf{x})$ detector resets 68 times after benign tilts and translations and never after a real drift, as Proposition 1 predicts. DRIFTTRIAGE is non-inferior to the DDM reset in accuracy (sine2d +0.17 points, better on 10 of 10 streams; lin −0.06 points), never resets after a coverage drift, makes 9 instead of 47 needless resets, and keeps the confidence-based detection of a new class outside the support (AUROC

0.786 versus 0.606, Wilcoxon $p = 0.001$). It does not improve on DDM in accuracy, and when a new class appears inside the support (lin), the new class is also a real drift for the region it takes over; both reset policies then lose novelty detection (0.66 and 0.64 versus 0.87 for FIFO). All four pre-registered stream hypotheses held; of the observation checks, the bound on the accuracy loss under in-support tilt was missed by 0.0006 (Observation 1).

6. Related Work

Concept drift (Gama et al., 2014; Lu et al., 2019); concept drift for new model families, e.g. multi-modal LLMs (Yang et al., 2025a;b; 2026); TFM under shift (Helli et al., 2024; Lourenço et al., 2026). **Difference to the competitors.** Drift-Resilient TabPFN retrains the prior to extrapolate temporal shift and Lourenço et al. (2026) manage the context with a fixed FIFO memory; both target accuracy after adaptation and treat drift as one phenomenon. DRIFTTRIAGE is training-free and decides the drift type before acting, including the rejection of unseen classes.

A. Design History and Pre-Registration

All choices were made on development seeds 0–4 and frozen before the single run on held-out seeds 10–19. **v0** compared the current context with a context of the two previous batches on the newest labelled batch (context-swap test); it detected real drift about three batches late, because the comparison context still held pre-drift rows. **v1** cross-fitted the newest batch instead; with 50 rows per fold it could not learn complex boundaries and missed real drift (0 resets on one stream). **v2** (the evaluated method) filters the error stream of a standard DDM detector. Two refinements were rejected on development streams: counting a class as known only after 30 context rows (cascades of resets after every reset) and a refractory period with local pruning of rows taken over by a new class (lower accuracy; the refractory period also made DDM worse). The hypotheses of Section 5, including the observation checks, were written down before the held-out run; all four stream hypotheses held and one observation bound was missed by 0.0006.

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